Per-site ΔRMSE from adding disturbance — M3 (C+T) vs M4 (C+T+D)

Below 0 = D reduces the site's RMSE | bars coloured by relative mortality (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (102 sites, 362 site-years) · deadtrees v4.1

XGBoost (Optuna) · 12m | **XGBoost (Optuna) · 24m** | **LightGBM (Optuna) · 12m** | **LightGBM (Optuna) · 24m**

GPP (g C m⁻² season⁻¹)

59% improved | 63% improved | 60% improved | 66% improved

200 | 0 | -200

NEP (g C m⁻² season⁻¹)

59% improved | 64% improved | 60% improved | 61% improved

100 | 0 | -100

ET (mm season⁻¹)

38% improved | 45% improved | 52% improved | 47% improved

40 | 0 | -40

WUE (g C mm⁻¹)

55% improved | 66% improved | 60% improved | 61% improved

0.5 | 0.0 | -0.5 | -1.0

ΔRMSE = RMSE(with D) − RMSE(without D)

Sites (sorted by ΔRMSE within each panel)

Relative mortality (%)
site maximum, 500 m
breaks 9.2 | 18.7

50
40
30
20
10
0